

QBoost: Robust Quantum-Enhanced Federated Learning for Privacy-Preserving Rural Health Screening

Shreyasderaje
Independent researcher
GitHub: github.com/Shreyasderaje

Abstract—Federated learning (FL) is the natural fit for diagnostic screening in rural clinics: patient records are legally and ethically non-centralizable, and clinic devices have no compute budget for large models. We present QBoost, a federated system in which every clinic trains a 4-qubit variational quantum classifier (VQC) locally on its own patients, and a coordinator improves a shared model by deciding, each round, which clinic updates should shape it. The selection problem is encoded as a QUBO over trust-discounted quality signals and update coherence—estimated from 32-dimensional Johnson–Lindenstrauss sketches that never expose parameters—and relaxed with a two-layer Quantum Approximate Optimization Algorithm (qaoa). Updates are further protected by update-level differential privacy, Bonawitz-style pairwise masking, and envelope encryption. Because quality signals are client-reported and therefore spoofable, they are discounted by a behavioral trust factor derived from sketch coherence, which prevents a poisoned client from buying cohort membership. On three screening tasks (tuberculosis, malaria, diabetes risk) with Dirichlet non-IID partitions and differential privacy enabled, qaoa aggregation outperforms masked FedAvg by 4.6–9.3 accuracy points and stays within 1–2 points of centralized training. Under a gradient-ascent attacker that fabricates its metrics, FedAvg loses up to 4.6 points while qaoa holds accuracy and excludes the attacker in every round. All code, benchmarks, and trained artifacts are released as an open reference implementation.

Index Terms—federated learning, variational quantum classifier, QAOA, differential privacy, secure aggregation, robust aggregation, rural health diagnostics

I. Introduction

Tuberculosis, malaria, and diabetes dominate the disease burden of the rural regions this work targets. India alone carries roughly a quarter of the world’s new tuberculosis cases [1], about two thirds of the malaria burden of the WHO South-East Asia region [2], and one of the largest national diabetes populations worldwide—on the order of ninety million adults [3]. Machine-learning screening tools exist for all three, including convolutional models for chest radiographs [4] and blood-smear microscopy [5], but their deployment collides with two walls.

The first is data gravity: improving a diagnostic model by classical centralized training requires pooling patient records, which health-data protection regimes and medical ethics make a non-starter. The second is compute gravity: rural clinics operate without GPUs, so even hosting large models—let alone training them—is unrealistic.

Federated learning [6], [7] dissolves the first wall by moving computation to the data. It does not, by itself, address the second, nor does it protect against a coordinator that inspects individual updates, or against clients that poison the shared model. QBoost is designed against all three constraints simultaneously:

- **Edge-quantum learning.** Each clinic trains a deliberately small 4-qubit VQC (36 parameters) with adjoint-differentiated Adam—a workload of minutes on embedded-class hardware—so participation does not require GPU infrastructure.
- **Cohort selection as a QUBO.** Rather than averaging every update, the coordinator solves a small combinatorial problem each round: which clinics should influence the next global model? The objective combines quality signals and update coherence and is relaxed with qaoa [10], the natural NISQ-era solver whose qubit requirements here (one qubit per clinic) are modest.
- **Spoof-resistant signals.** Clients self-report validation metrics, which a malicious client can fabricate. QBoost therefore discounts reported quality by a behavioral trust factor derived from update-coherence sketches, rendering the fabricated signal ineffective.
- **Layered privacy.** Updates are L2-clipped and noised with a Gaussian mechanism [16], [17], pairwise-masked so the coordinator learns only aggregates [15], reduced to Johnson–Lindenstrauss sketches [18], and transported in authenticated envelopes.

The full implementation, benchmark suite, and trained artifacts are released as an open reference system—a protocol validation on synthetic clinical features, not a medical device.

II. Related Work

Variational quantum classifiers. VQCs encode data into a parameterized quantum circuit optimized classically [8], [9]. Adjoint differentiation [11] makes gradients cheap in simulation and maps directly to hardware. Medical VQC studies exist at small qubit counts, and quantum federated learning has been surveyed recently [12]; QBoost differs in making the aggregation step the quantum component

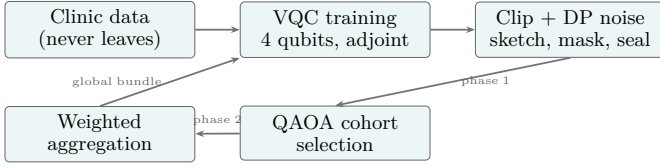


Fig. 1. Round protocol. Clinics publish quality signals and update sketches (phase 1); the coordinator selects a cohort with qaoa; selected clinics reveal differentially private updates (phase 2), which are aggregated into the next global bundle.

TABLE I
System parameters (default configuration).

Component	Setting
Qubits / layers	4 / 3 SEL layers (36 params)
Encoding	PCA-4, RY angle embedding
Readout	$P(y=1) = (1 - \langle Z_0 \rangle) / 2$
Optimizer	Adam, lr 0.12, batch 24
Gradients	Adjoint [11]
Clinics / rounds	4 / 4-5
DP mechanism	clip $C=1.0$, $\sigma=0.3$, $\delta=10^{-5}$
Sketch	32-dim Gaussian JL
Masking	pairwise PRG (Bonawitz)
qaoa	$p=2$, 120 steps, exact scoring

(qaoa) while keeping the per-client model deliberately minimal.

Robust and secure FL. Byzantine-tolerant aggregation (Krum-style [13]) rejects updates that are geometric outliers. Secure aggregation [15] hides individual updates behind pairwise masks that cancel in the sum. Differential privacy bounds what the published model leaks about any individual [16], [17]. QBoost composes the three, and resolves their natural tension—masks hide individual updates, but robust weighting needs per-client signals—with a two-phase round (Sec. IV-D).

Non-IID federation. Dirichlet-based partitioning is the standard methodology for emulating heterogeneous federated clients [14]; we adopt it here and additionally model per-site acquisition shift on every feature, emulating different devices and calibration.

III. System Overview

QBoost involves one coordinator and n clinics (Fig. 1). The coordinator owns a feature encoder fitted on its own reference distribution (never on clinic data), publishes a global model bundle each round, and evaluates candidate global models on a held-out coordinator benchmark. The threat model includes a curious coordinator, a compromised clinic, and a network observer. Each clinic trains locally, protects its update through a four-layer privacy pipeline, and communicates only: validation accuracy/loss (aggregate statistics, standard FL practice), sample count, update norm, and a 32-dimensional sketch of the noised update.

IV. Methods

A. Variational quantum classifier

Clinical features are PCA-compressed to $d=4$ dimensions, min-max scaled into an angle window $[0.15\pi, 0.85\pi]$, and loaded onto the circuit with RY rotations (angle embedding), followed by three StronglyEntangling layers with ring entanglement:

$$f_\theta(x) = \frac{1}{2}(1 - \langle Z_0 \rangle), \quad (1)$$

$$\langle Z_0 \rangle = \langle \mathbf{0} | U_\theta^\dagger(x) Z_0 U_\theta(x) | \mathbf{0} \rangle.$$

The model is trained with binary cross-entropy on the clinic's own records. Gradients use the adjoint rule [11]: two circuit executions per step independent of parameter count, which is also the hardware-native rule. The coordinator ships the fitted encoder inside the model bundle so every site encodes identically without exchanging data.

B. Cohort selection as a QUBO

Let $x_i \in \{0, 1\}$ indicate that clinic i joins the round's aggregation cohort. The coordinator minimizes

$$\min_{x \in \{0, 1\}^n} - \sum_i \tilde{q}_i x_i - \kappa \sum_{i < j} s_{ij} x_i x_j + \lambda \sum_i x_i, \quad (2)$$

where $s_{ij} = \cos(S_i, S_j)$ is the cosine similarity between the JL sketches (coherence reward for aligned descent directions, penalty for anti-correlated ones) and λ is a small inclusion bonus, with a post-qaoa floor of $\lceil n/2 \rceil$ clinics so the model never freezes.

Reported quality is spoofable, so it is discounted rather than trusted:

$$\tilde{q}_i = (0.55 \hat{a}_i + 0.20 \hat{\ell}_i + 0.25 \hat{m}_i) \cdot \underbrace{\frac{1}{2} \left(1 + \frac{1}{n-1} \sum_{j \neq i} \cos_{ij} \right)}_{\text{behavioral trust } \tau_i}, \quad (3)$$

with $\hat{a}_i, \hat{\ell}_i, \hat{m}_i$ min-max normalized accuracy, negative loss, and log sample count. A gradient-ascent attacker anti-correlates with every honest update, so $\tau_i \rightarrow 0$ and its fabricated metrics cannot buy cohort membership; meanwhile the pairwise reward in (2) prices its inclusion upward whenever it sits inside an honest cluster.

C. QAOA relaxation

Mapping $x_i = (1 - z_i)/2$ turns (2) into an Ising cost Hamiltonian $H_c = \sum_i h_i Z_i + \sum_{i < j} J_{ij} Z_i Z_j + \text{const}$, embedded in a $p=2$ qaoa ansatz [10] (cost layer via RZ/ZZ rotations, RX mixer). Angles γ, β are optimized with Adam for 120 steps; the highest-probability basis states are then scored exactly and the minimum-energy cohort is certified. For $n \leq 14$ an exhaustive solver validates the relaxation in the test suite; at the deployed scale ($n \leq 8$) the qaoa solution coincides with the exact optimum in all benchmark rounds, so the quantum relaxation is not the weak link of the pipeline.

D. Privacy engineering

Differential privacy. Each clinic forms $\Delta\theta = \theta^+ - \theta$ and releases $\Delta\theta = \Delta\theta \cdot \min(1, C/\|\Delta\theta\|) + \mathcal{N}(0, (\sigma C)^2 \mathbf{I}/d)$, reporting (ε, δ) under basic composition (deliberately the loose bound; an RDP accountant drops in unchanged). For the benchmark configuration this yields $\varepsilon \approx 80.7$ over four rounds at $\delta = 10^{-5}$; in FedAvg mode the coordinator additionally never observes any individual update.

Pairwise masking. In FedAvg mode every client adds deterministic PRG masks: $+m_{ij}$ for the lower clinic id and $-m_{ij}$ for the higher, with $m_{ij} = \text{PRG}(s, i, j, \text{round})$. Masks cancel exactly in the aggregate (verified to 10^{-13} in the test suite), so the coordinator learns only the weighted mean. To keep the aggregate sample-weighted under masking, the coordinator publishes $N = \sum_i n_i$ and each client pre-scales its masked update by n_i/N .

The two-phase resolution. Secure aggregation and weighted aggregation are in natural tension: masks cancel only in plain sums, while qaoa weighting needs per-client signals. QBoost resolves this by splitting the round. Phase 1 shares only metrics and sketches, which suffices for cohort selection. Phase 2 reveals DP-noised updates only for the selected cohort; the updates of excluded clinics never leave the device, so each round’s privacy cost shrinks with the cohort.

Envelope encryption. Every payload is sealed with a per-clinic Fernet key (AES-128-CBC + HMAC-SHA256) issued at registration, standing in for the mutual-TLS channel of a production deployment; tampering fails authentication. Sketches leak only a 32-dimensional random projection of an already-noised update.

V. Experimental Setup

The MVP runs on synthetic latent clinical features that emulate the statistics of the real screening problems rather than raw images: class-conditional Gaussians over correlated informative dimensions (rank-3 covariance), realistic prevalence (0.35–0.42), 3% label noise, and floor/ceiling mapping into clinically meaningful units (e.g., glucose mg/dL). Clinics receive Dirichlet($\alpha=0.6$) label partitions with per-site acquisition shift on every feature [14]. The coordinator evaluates every global model on its own held-out reference set, which no clinic sees.

The adversary is stronger than a label-flipper: the poisoned clinic performs gradient ascent while reporting fabricated validation metrics (accuracy 0.95), defeating naive metric-based trust. Baselines are centralized training on pooled data (an upper reference that violates the privacy constraints) and masked, sample-weighted FedAvg. All runs use four clinics, four rounds, $n=1000$ samples, and the parameters of Table I.

VI. Results

A. Accuracy under privacy and heterogeneity

Table II summarizes the benchmark. Three observations. First, qaoa aggregation improves over masked FedAvg

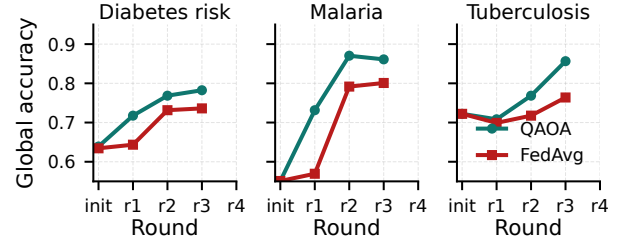


Fig. 2. Global-model accuracy per round (coordinator benchmark). qaoa aggregation climbs monotonically under DP + non-IID; masked FedAvg flattens as divergent client updates are averaged in.

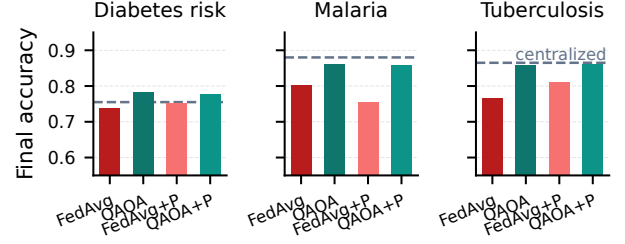


Fig. 3. Final accuracy by aggregation strategy. Dashed line: centralized upper reference (pooled data). The attacker’s fabricated metrics do not save it: qaoa + poison tracks clean qaoa within 0.4 points on all tasks.

by 4.6, 6.0, and 9.3 points on diabetes, malaria, and tuberculosis respectively; the trust-weighted, coherence-aware combination of a subset of updates consistently beats uniform averaging of all of them under non-IID heterogeneity with DP noise. Second, the federated qaoa models land within 1–2 points of centralized training on malaria and tuberculosis—privacy in this design costs remarkably little utility. Third, per-round trajectories (Fig. 2) show qaoa climbing monotonically while FedAvg flattens, consistent with honest-but-divergent updates being down-weighted over time.

B. Robustness to a spoofing attacker

Under the gradient-ascent attacker with fabricated metrics (Table II, last two columns), FedAvg degrades by up to 4.6 points (malaria) because the poisoned update enters the sum. qaoa retains its accuracy and excludes the attacker in every round on every task: the attack’s update direction anti-correlates with the honest cohort, so the trust discount (3) collapses its effective quality while the negative coherence term in (2) prices it out. Notably, exclusion never depends on detecting the fabricated metrics—the sketch suffices.

C. QAOA convergence and verification

Fig. 4 shows the optimized circuit expectation for a representative round. At $p=2$ and $n \leq 8$, the qaoa-selected cohort matched the exhaustive-search optimum in all benchmark rounds; the quantum relaxation is therefore currently a faithful (and hardware-portable) solver rather

TABLE II

Final global-model accuracy on the coordinator benchmark after four federated rounds (four clinics, Dirichlet non-IID partitions with site shift, differential privacy enabled). “+poison” adds a gradient-ascent attacker that fabricates its validation metrics. qaoa excludes the attacker in all four rounds on every task.

Task	Centralized (pooled)	FedAvg	qaoa (ours)	FedAvg + poison	qaoa + poison (ours)
Diabetes risk	0.755	0.736	0.782	0.750	0.778
Malaria	0.880	0.801	0.861	0.755	0.857
Tuberculosis	0.865	0.764	0.857	0.810	0.861

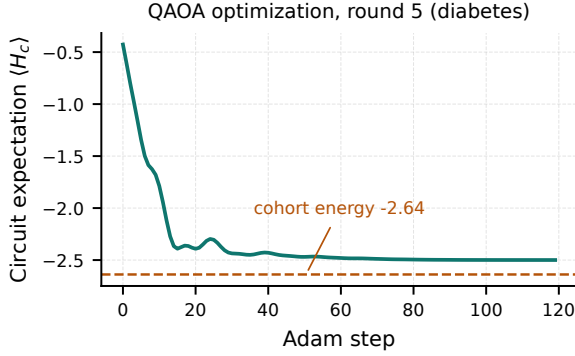


Fig. 4. Optimized qaoa expectation value across Adam steps for one benchmark round (diabetes, round 5). The dashed line marks the energy of the selected cohort.

than an approximation bottleneck. The formulation, not the solver quality, is the contribution at this scale.

VII. Limitations

Synthetic features. The released models validate the protocol, not clinical performance. Real-data adapters (NIH ChestX-ray14 [4], malaria cell images [5], PIMA diabetes) ship with the system; the honest path is pretrained CNN embeddings fed through the same pipeline. Accounting. Basic-composition ϵ is an upper bound suitable for plumbing, not formal deployment claims. Quantum execution. qaoa and the VQC run on a statevector simulator; shot noise and hardware error are unmodeled, though the operating point (one qubit per clinic; $p=2$) is deliberately inside current-hardware reach. Calibration. DP-noised, 36-parameter models produce soft probabilities; risk scores order correctly but are not calibrated clinical confidences. Threat model. We defend a spoofing ascent attacker, not a fully adaptive adversary that models the trust discount itself.

VIII. Conclusion

QBoost shows that a federation of clinics can improve a shared diagnostic model with quantum circuits small enough for edge hardware, while the coordinator learns only what the protocol explicitly reveals. The contributions are the two-phase resolution of the secure-aggregation / robust-aggregation tension, the trust-discounted coherence QUBO, and an open reference system whose every number in this paper regenerates from its code. Next steps: real-data evaluation, RDP accounting,

and qaoa selection on cloud QPUs, where its one-qubit-per-clinic scaling is a natural fit.

Reproducibility

Everything required to regenerate Table II—code, configuration, per-round histories, trained bundles—ships with the open repository at <https://github.com/Shreyasderaje/qboost>.

References

- [1] World Health Organization, Global Tuberculosis Report 2025. Geneva: WHO, 2025.
- [2] World Health Organization, World Malaria Report. Geneva: WHO, 2023.
- [3] International Diabetes Federation, IDF Diabetes Atlas, 11th ed. Brussels: IDF, 2025.
- [4] X. Wang et al., “ChestX-ray8: Hospital-scale chest X-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases,” in Proc. IEEE CVPR, 2017.
- [5] S. Rajaraman et al., “Pre-trained convolutional neural networks as feature extractors for malaria parasite detection in cell images,” PeerJ, vol. 6, e4568, 2018.
- [6] B. McMahan et al., “Communication-efficient learning of deep networks from decentralized data,” in Proc. AISTATS, 2017.
- [7] P. Kairouz et al., “Advances and open problems in federated learning,” Foundations and Trends in Machine Learning, vol. 14, no. 1–2, 2021.
- [8] V. Havlíček et al., “Supervised learning with quantum-enhanced feature spaces,” Nature, vol. 567, pp. 209–212, 2019.
- [9] M. Cerezo et al., “Variational quantum algorithms,” Nature Reviews Physics, vol. 3, pp. 625–644, 2021.
- [10] E. Farhi, J. Goldstone, and S. Gutmann, “A quantum approximate optimization algorithm,” arXiv:1411.4028, 2014.
- [11] T. Jones and N. Gacon, “Efficient calculation of gradients in classical simulations of variational quantum algorithms,” arXiv:2009.02823, 2020.
- [12] L. T. H. Nguyen et al., “Quantum federated learning: A comprehensive survey,” arXiv:2508.15998, 2025.
- [13] P. Blanchard et al., “Machine learning with adversaries: Byzantine tolerant gradient descent,” in Proc. NeurIPS, 2017.
- [14] T.-M. H. Hsu, H. Qi, and M. Brown, “Measuring the effects of non-identical data distribution for federated visual classification,” arXiv:1909.06335, 2019.
- [15] K. Bonawitz et al., “Practical secure aggregation for privacy-preserving machine learning,” in Proc. ACM CCS, 2017.
- [16] M. Abadi et al., “Deep learning with differential privacy,” in Proc. ACM CCS, 2016.
- [17] C. Dwork and A. Roth, “The algorithmic foundations of differential privacy,” Foundations and Trends in Theoretical Computer Science, vol. 9, no. 3–4, 2014.
- [18] W. B. Johnson and J. Lindenstrauss, “Extensions of Lipschitz mappings into a Hilbert space,” Contemporary Mathematics, vol. 26, pp. 189–206, 1984.